

UTKALIKA SATAPATHY

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Education

- Aug 2021– May 2026 **Indian Institute of Technology Kharagpur, West Bengal, India, Ph.D., CSE,**
Supervisor: Prof. Sandip Chakraborty
Research Area: Operating Systems, Distributed Systems, and Observability
Research Group: UbiNet Lab
Thesis: Provenance and Observability Frameworks for Secure Distributed Microservices.
- 2019–2017 **IIIT Bhubaneswar, Odisha, India, M.Tech., CSE,**
CGPA: 8.66/10
Supervisor: Prof. Debashis Jena
Research Group: ISEA Lab.
- 2013–2009 **Silicon Institute of Technology, Bhubaneswar, India, B.Tech., IT,**
CGPA: 8.46/10 .
- 2009–2007 **Rama Devi Women's Junior College, Bhubaneswar, Higher Secondary.**

Employment/Internship

- Aug 2025 – Present **Assistant Professor (Contractual), IIIT Bhubaneswar, Odisha,**
Courses Teaching: Operating System, Object Oriented Programming using C++, Database Management Systems.
- May 2023 – Aug 2023 **Summer Research Intern, IBM Research Lab, Bangalore,**
Project Title: Policy Enforcement In eBPF based Cloud infrastructure
Project Description: Developed a Linux Security Module (LSM) to enforce policy-based access control and security checks for eBPF programs, ensuring compliance with defined guidelines in cloud infrastructure..
- Feb 2023 – Present **Senior Research Fellow, IIT Kharagpur sponsored by IGCAR India,**
Project Title: Development of Algorithms and Tools for Log Analytics
Project Description: To design a unified observability framework for robust, scalable, and efficient logging, provenance tracking, and attack detection in distributed microservice-based applications..
- Feb 2021 – Feb 2023 **Junior Research Fellow, IIT Kharagpur sponsored by IGCAR India,**
Project Title: Development of Algorithms and Tools for Log Analytics
Project Description: To design a unified observability framework for robust, scalable, and efficient logging, provenance tracking, and attack detection in distributed microservice-based applications..
- Sep 2019 – Jan 2021 **Assistant Professor (Contractual), CET, Odisha (Now OUTR, Odisha),**
Courses Teaching: Programming for Problem Solving, Java, Data Structure, Internet and Web Technology.
- May 2018 – Jun 2018 **Summer Research Intern, IIT Kharagpur,**
Project Title: SilentInformer: A framework for Information Dispersion using Inaudible Acoustic Signals
Project Description: Developed a smartphone-based system for information sharing via inaudible acoustic signals, achieving low bit error rates ($BER \leq 10\%$) in real-world noisy environments..
- Sep 2013 – Jul 2017 **Senior System Engineer, Infosys Ltd., Java Developer, at AMEX Account.**

Awards & Grants

- 2026 **COMSNETS Excellence in Review Award**, for revieweing Graduate Forum of COMSNETS 2026.
- 2025 **COMSNETS Best Paper 1st Runner Up Award**, for presenting the paper titled "Towards Generating a Robust, Scalable and Dynamic Provenance Graph for Attack Investigation over Distributed Microservice Architecture".
- 2025 **COMSNETS Travel Grant**, for attending COMSNETS 2025.
- 2024 **Google Research India**, for participating in Research Week with Google 2024.
- 2024 **ACM India ARCS and Annual Event**, for presenting the paper titled DisProTrack as a part of the Lightning Talks and Posters session.
- 2023 **LRN Foundation International Travel Grant Award in Association with COMSNETS**, for attending IEEE INFOCOM 2023.
- 2023 **ACM-India and IARCS Travel Grant**, for attending IEEE INFOCOM 2023.
- 2023 **INFOCOM 2023 Conference Travel Grant**, for attending IEEE INFOCOM 2023.
- 2018 **Mobicom'18 Scholarship Grant**, for attending ACM Mobicom 2018.
- 2017 **GATE**, Qualified, *Score:432/1000*.

Selected Publications

- IEEE TSC Journal 2025 **XPLOG: A Dynamic Observability Framework for Distributed Sandboxed Microservices**,
Utkalika Satapathy, Harsh Borse, Rajat Bachhawat, Neha Dalmia, Subhrendu Chattopadhyay, Sandip Chakraborty, in IEEE Transactions on Services Computing, 2025.
- IEEE TSC Journal 2025 (Submitted) **μ ProvGAE: Anomaly Detection with Enriched Provenance Graphs and GAE in Distributed Microservice Architecture**,
Utkalika Satapathy, Harsh Borse, Sandip Chakraborty, in IEEE Transactions on Services Computing, 2025.
- COMSNETS 2025 **Towards Generating a Robust, Scalable and Dynamic Provenance Graph for Attack Investigation over Distributed Microservice Architecture**,
Utkalika Satapathy, Harsh Borse, Sandip Chakraborty, in International Conference on Communication Systems & NETWORKS (COMSNETS), 2024.
- COMSNETS 2025 **BeeGuard: Explainability-based Policy Enforcement of eBPF Codes for Cloud-native Environments**,
Neha Chowdhary, Utkalika Satapathy, T A. Benson, S Chattopadhyay, P Kodeswaran, S Sen, S Chakraborty, in International Conference on Communication Systems & NETWORKS (COMSNETS), 2024.
- IEEE ICDCS POSTER 2024 **URCD: Unsupervised Root Cause Detection in Microservices Architecture with HGAN**,
Harsh Borse, Utkalika Satapathy, Mainak Mondal, Bivas Mitra, IEEE International Conference on Distributed Computing Systems (Posters and Demos), 2024.
- IEEE INFOCOM 2023 **DisProTrack: Distributed Provenance Tracking over Serverless Applications**,
Utkalika Satapathy, Rishabh Thakur, Subhrendu Chattopadhyay, Sandip Chakraborty, IEEE International Conference on Computer Communications (INFOCOM), 2023.
- IJIVICS Journal 2022 **Collaborative decision making system in intelligent transportation system using distributed blockchain technology**,
Bhabendu Kumar Mohanta, Debasish Jena, Utkalika Satapathy, Somula Ramasubbareddy, International Journal of Vehicle Information and Communication Systems, 2022.
- COMSNETS 2020 **SilentInformer: A Framework for Information Dispersion Using Inaudible Acoustic Signals**,
Utkalika Satapathy, Soumyajit Chatterjee, Sandip Chakraborty, in International Conference on Communication Systems & NETWORKS (COMSNETS), 2020.

- Elsevier IoT Journal 2020 **Survey on IoT security: Challenges and solution using machine learning, artificial intelligence and blockchain technology**,
Bhabendu Kumar Mohanta, Debasish Jena, Utkalika Satapathy, Srikanta Patnaik, Internet of Things; Engineering Cyber Physical Human Systems, 2020.
- ICIIS 2018 **An ECC-based Lightweight Authentication protocol for Mobile Phone in Smart home**,
Utkalika Satapathy, Bhabendu Ku. Mohanta, Debasish Jena, Srichandan Shobhanayak, in IEEE 13th International Conference on Industrial and Information Systems (ICIIS) at IIT Ropar, 2018.
- ICCCNT 2019 **Secure Communication in Internet of Things Application using Hyperledger based Blockchain**,
Utkalika Satapathy, Bhabendu Ku. Mohanta, Soumyashree S. Panda, Srichandan Shobhanayak, Debasish Jena, in IEEE 10th International Conference on Computing, Communication And Networking Technologies (ICCCNT) at IIT Kanpur, 2019.
- IJSTR 2019 **Trust Management in IoT Enable Healthcare System using Ethereum based Smart Contract**,
Utkalika Satapathy, Bhabendu Ku. Mohanta, Debasish Jena.

Teaching & Lab

- Jan 2026 **Database Management Systems Theory**, Dept. of CSE, IIIT Bhubaneswar, Odisha.
<https://usatpath01.github.io/courses/2026-spring-DBMS>
- Jan 2026 **Database Management Systems Lab**, Dept. of CSE, IIIT Bhubaneswar, Odisha.
<https://usatpath01.github.io/courses/2026-spring-DBMS>
- Aug 2025 **Object Oriented Programming Theory**, Dept. of CSE, IIIT Bhubaneswar, Odisha.
<https://usatpath01.github.io/courses/2025-autumn-OOP>
- Aug 2025 **Operating systems**, Dept. of CSE, IIIT Bhubaneswar, Odisha.
<https://usatpath01.github.io/courses/2025-autumn-OS>
- Aug 2020 **Java Programming Theory and Lab, Programming and Problem Solving Theory and Lab**,
Dept. of IT, OUTR, Odisha.
- Aug 2020 **Data Structures and Algorithms Theory, Internet and Web Technologies Theory**, Dept. of
IT, OUTR, Odisha.
- Aug 2019 **Programming and Problem Solving Theory and Lab, Java Programming Theory and Lab**,
Dept. of IT, OUTR, Odisha.

Teaching Assistance

- Autumn 2024-2025 **CS61065: Theory and Applications of Blockchain**, By Prof. Sandip Chakraborty & Prof. Shamik Sural, Dept. of CSE, IIT Kharagpur.
https://onlinecourses.nptel.ac.in/noc24_cs15/preview
- Jan–Apr 2023 **Theory of Blockchain & Application**, By Prof. Sandip Chakraborty & Prof. Shamik Sural, NPTEL Online Certification Course, IIT Kharagpur Dept. of CSE.
https://onlinecourses.nptel.ac.in/noc23_cs47/preview
- Spring 2023-2024 **CS19001: Programming & Data Structure Tutorial and Lab**, By Prof. Rajib Mall & Prof. Sudeshna Sarkar, Dept. of CSE, IIT Kharagpur.
<https://cse.iitkgp.ac.in/~aritrah/course/theory/PDS/Spring2024/>
- Autumn 2023-2024 **CS61065: Theory and Applications of Blockchain**, By Prof. Sandip Chakraborty, Dept. of CSE, IIT Kharagpur.
<https://sandipc-iitkgp.github.io/sandip-web/courses/cs61065/cs61065.html>
- Jan–Apr 2023 **Theory of Blockchain & Application**, By Prof. Sandip Chakraborty & Prof. Shamik Sural, NPTEL Online Certification Course, IIT Kharagpur Dept. of CSE.
https://onlinecourses.nptel.ac.in/noc23_cs47/preview

- Spring 2022-2023 **CS60002: Distributed Systems**, By Prof. Sandip Chakraborty, Dept. of CSE, IIT Kharagpur.
<https://cse.iitkgp.ac.in/~sandipc/courses/cs60002/cs60002.html>
- Autumn 2022-2023 **CS61065: Theory and Applications of Blockchain**, By Prof. Sandip Chakraborty, Dept. of CSE, IIT Kharagpur.
<https://sandipc-iitkgp.github.io/sandip-web/courses/cs61065/cs61065.html>
- Spring 2021-2022 **CS10003: Programming and Data Structure Theory & Lab**, By Prof. Sudebkumar Prasant Pal, Dept. of CSE, IIT Kharagpur.
<https://cse.iitkgp.ac.in/%7Esp/teaching.html>
- Jan–Apr 2022 **Theory of Blockchain & Application**, By Prof. Sandip Chakraborty & Prof. Shamik Sural, NPTEL Online Certification Course, IIT Kharagpur Dept. of CSE.
https://onlinecourses.nptel.ac.in/noc22_cs44/preview
- Spring 2019 **Operating System**, Dept. of CSE, IIIT Bhubaneswar.
- Aut 2018 **Java Programming**, Dept. of CSE, IIIT Bhubaneswar.
- Aut 2018 **Programming in C**, Dept. of CSE, IIIT Bhubaneswar.

Invited Talks

- Feb 2026 **Invited Talk**, at Faculty Updation Program (FUP) on Evolution of Cybersecurity: From Traditional Models to Quantum-Ready Cryptography held at IIIT Bhubaneswar during Feb 10-14, 2026
Title: Attack Investigation in Microservices Using Provenance Graphs.

Professional Responsibilities

- 2026 **Peer Reviewer**, for The 18th Annual International Conference on Communication Systems & Networks (COMSNETS 2026) Graduate Forum.
- 2026 **Program Co-ordinator**, for 2-Days Short Term Program on Recent Trends in AI and Cyber Security Organized by IIIT Bhubaneswar under ISEA Phase-III Project, <https://www.iiit-bh.ac.in/workshop/>.
- 2026 **Organizing Committee**, for 1st National Workshop On Sensor to Cloud: IoT Applications And Future Trends (SCIoT - 2026) Organized by Department of Computer Science, IIIT Bhubaneswar, <https://www.iiit-bh.ac.in/workshop/>.
- 2026 **Program Co-ordinator**, for One-Week Faculty Updation Program on Evolution of Cybersecurity: From Traditional Models to Quantum-Ready Cryptography Organized by IIIT Bhubaneswar under ISEA Phase-III Project, <https://www.iiit-bh.ac.in/workshop/>.
- 2026 **Program Co-ordinator**, for One-Week Bootcamp on Cybersecurity and Information Security in Industry 4.0 with Cryptology, Machine Learning and Federated Learning Organized by IIIT Bhubaneswar under ISEA Phase-III Project, <https://www.iiit-bh.ac.in/workshop/>.
- 2025 **Peer Reviewer**, for IEEE Transactions on Dependable and Secure Computing 2025 (TDSC 2025).
- 2025 **Peer Reviewer**, for IEEE Transactions on Services Computing 2025 (TSC 2025).
- 2025 **Organizing Committee**, for 18th National Workshop On Recent Trends in Software Testing (RTST - 2025) Organized by Department of Computer Science, IIIT Bhubaneswar, <https://www.iiit-bh.ac.in/workshop/>.
- 2025 **Organizing Committee**, for Indocrypt 2025, The 26th International Conference on Cryptology in India, <https://www.indocrypt2025.in/>.
- 2025 **Organizing Committee**, for InGARSS 2025, for India Geoscience and Remote Sensing Symposium, <https://www.ingarss-2025.in/>.
- 2025 **Organizing Committee**, for ISEA National Workshop 2025, CipherStorm: Unlocking Digital Security organized by IIIT Bhubaneswar in collaboration with ISI Kolkata under ISEA Phase-III Project, <https://isea.gov.in/event.php/IVP20251031JASE>.
- 2024 **Peer Reviewer**, for IEEE Transactions on Services Computing 2024 (TSC 2024).

- 2023 **Volunteer**, for ICDCN 2023, The 24th International Conference On Distributed Computing And Networking, <https://cse.iitkgp.ac.in/conf/ICDCN23/>.
- 2022 **Web Chair**, for AI-ML Systems 2022, The Second International Conference on AI-ML Systems, <https://www.aimlsystems.org/2022/>.
- 2021 **Volunteer**, for AI-ML Systems 2021, The First International Conference on AI-ML Systems, <https://www.aimlsystems.org/2023/>.
- 2020 **Peer Reviewer**, for IEEE internet of things journal.

Research Projects

- Apr 2024 **Leveraging Micro-resource Observation for Non-invasive Trace Analysis for Distributed Applications**, Developed a trace aggregation technique and automated tool to hierarchically analyze execution traces, enabling precise identification of performance bottlenecks and enhancing diagnosis accuracy..
- Apr 2024 **Enforcing Organizational Policies as Security Rules Within the System**, Designed a dynamic policy enforcement framework integrating Open Policy Agent (OPA) with eBPF at the kernel level, enabling low-level network packet inspection and automated conversion of human-readable security policies into machine-executable policies using advanced AI techniques..
- Nov 2023 **Unified System and Application Log Generation and Analysis on Serverless Environment using eBPF**, Developed a web-based interface for parsing eBPF-generated unified logs, enabling intuitive visualization, efficient debugging, and enhanced security in containerized computing environments..
- Nov 2023 **Attack Investigation in Distributed Environments**, Designed a tailored attack investigation system for distributed systems to enhance detection, response, and defense against Advanced Persistent Threat (APT) attacks..
- Nov 2022 **Reverse Engineering on Docker Images**, Proposed a method to extract source code or binaries from containerized serverless applications to enable effective provenance tracking and issue diagnosis..
- Nov 2022 **Unified System Call Log and Application Log Generation on Serverless Environment using eBPF**, Developed a novel eBPF-based tool to generate unified system and application logs, enabling efficient provenance tracking across containerized applications and multiple hosts for root cause analysis..

Technical Skills

Programming	Python, C, C++, Java, SQL
Platforms & Tools	eBPF, Linux Kernel Module, Git, Docker, Virtual Box, MATLAB, TEXnicCenter, Anaconda Navigator, Google Colab, Pycharm, Eclipse IDE, Jenkins, MySQL Workbench
Databases	Oracle 10g, DB2, MySQL, MongoDB
Operating System	Ubuntu, Mac, Windows 10, Kali Linux

Extra-Curricular Activities

Languages	English, Hindi, Odia
Dance	Awarded with first prize in Infosys Got Talent (IGT) 2.0 in Infosys, 1st Runner-up in Infosys Dancing Star Competition

Personal Details

Date of Birth	01-04-1992
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References

- 1 **Prof. Sandip Chakraborty**, *Associate Professor, Department of Computer Science and Engineering, Indian Institute of Technology Kharagpur, India,*
Email: sandipc@cse.iitkgp.ac.in
Phone: (+91) (3222) (282898)
Relationship: PhD Supervisor.
- 2 **Prof. Debashish Jena**, *Associate Professor, Department of Computer Science, IIIT Bhubaneswar, India,*
Email: debasish@iiit-bh.ac.in
Phone: (0674) (3060519)
Relationship: M.Tech Supervisor.